



BIG DATA?

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WHAT IS BIG DATA?

- Extremely large datasets
- That are too complex
- Too voluminous to be processed analyzed using traditional data-processing techniques



BIG DATA OVERVIEW

Big Data is a collection of data that is huge in volume, yet growing exponentially with time.

- Sources of Big Data

- Social media platforms (e.g., Facebook, Twitter)
- Sensor networks (e.g., IoT devices)
- Transactional data (e.g., e-commerce transactions)
- Log files (e.g., server logs)
- Medical records and genomic data

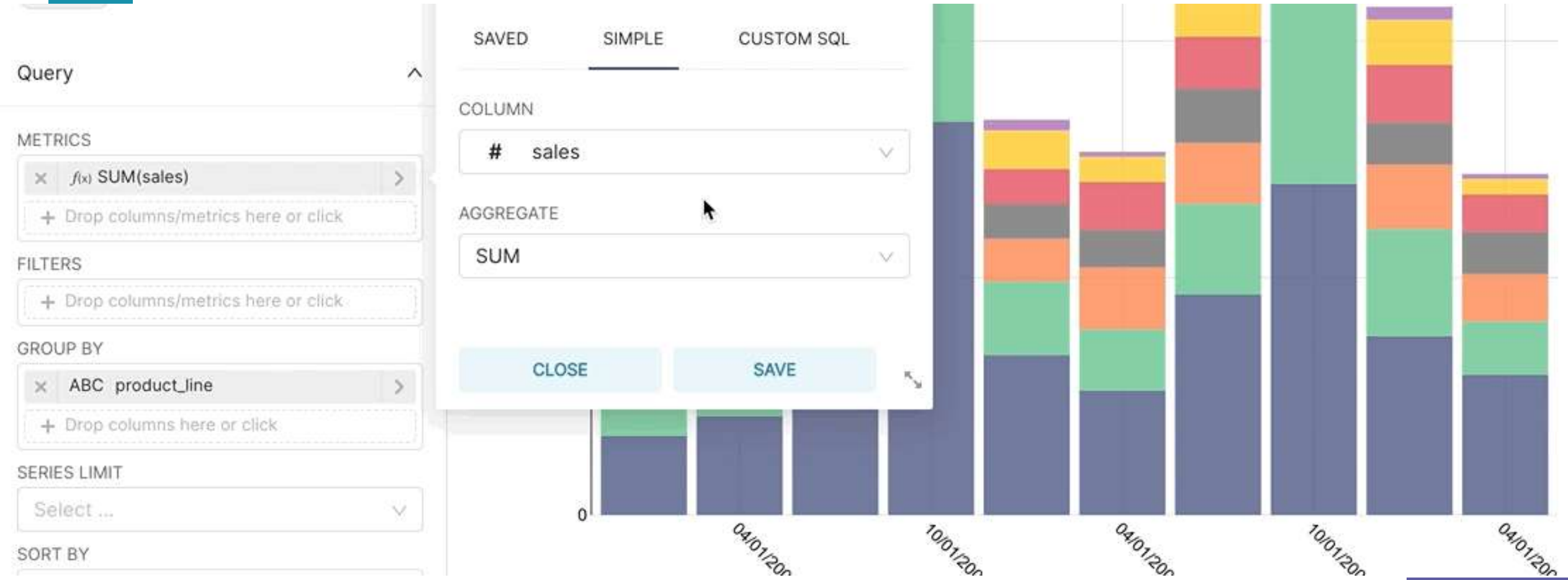


BUSINESS INTELLIGENCE

- Uses Data To Provide Insight Into The Business Records
- It helps the team to evaluate the journey efficiently and gauge the direction where the company is heading towards



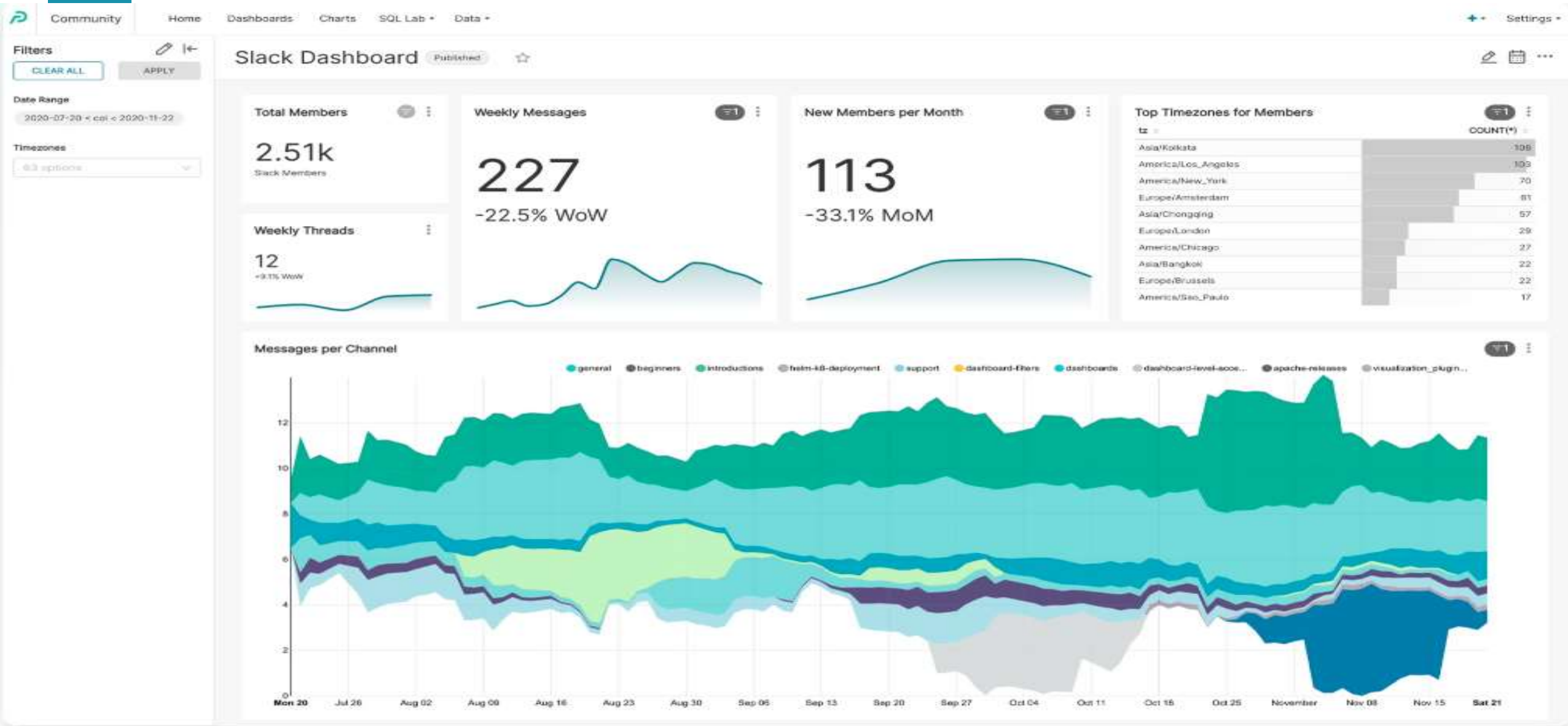
BUSINESS INTELLIGENCE TOOL EXAMPLE



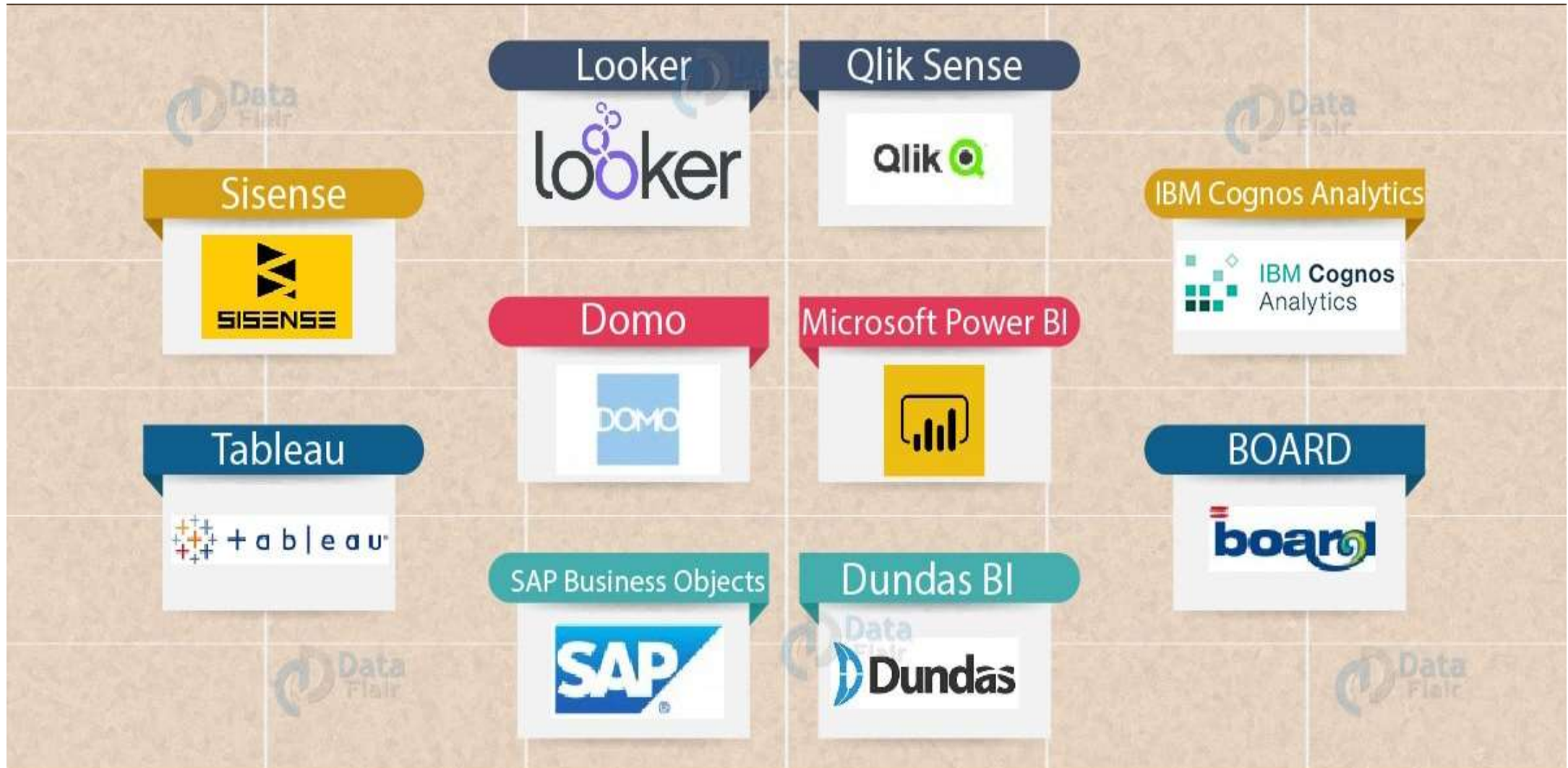
BUSINESS INTELLIGENCE TOOL EXAMPLE



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BUSINESS INTELLIGENCE Vs DATA ANALYTICS



BUSINESS INTELLIGENCE Vs DATA ANALYTICS

Business Intelligence	Data Analytics
Supports decision-making and business growth	Organizes, cleanses, predicts, and transforms data based on specific needs
Understands overall business operations	Answer specific questions, leading to a smaller scope for analytics within a business
Centers on descriptive data with a historical focus	Includes predictive and prescriptive analytics, looking into the future
Uses dedicated BI tools and historical data in data marts and warehouses	It relies on various data storage tools and BI tools depending on the strategy
BI and data analytics differ in reporting and data visualization based on scenarios and data types	Reporting is ideal for forecasting future trends using past data

Need of Data Analytics



NEED OF DATA ANALYTICS

Informed Decision-Making

- **Insight Generation:**

- Data analytics helps organizations uncover patterns, trends, and correlations in data, providing insights that drive strategic decisions.

- **Risk Management:**

- By analyzing historical data and identifying potential risks, organizations can mitigate uncertainties and plan for contingencies

NEED OF DATA ANALYTICS

Operational Efficiency

•Process Optimization:

- Analyzing operational data helps identify inefficiencies and bottlenecks, enabling process improvements and cost reductions.

•Resource Allocation:

- Data analytics assists in optimizing resource allocation by forecasting demand and managing inventory effectively.

NEED OF DATA ANALYTICS

Customer Understanding and Personalization

- **Customer Segmentation:**

- By analyzing customer data, organizations can segment their audience based on behavior, preferences, and demographics, leading to more targeted marketing efforts.

- **Personalized Experiences:**

- Data analytics allows for the creation of personalized customer experiences, enhancing customer satisfaction and loyalty.

NEED OF DATA ANALYTICS

. Product and Service Innovation

•Market Trends:

- Identifying emerging market trends through data analytics helps organizations innovate and develop new products or services.

•Feedback Analysis:

- Analyzing customer feedback and usage data enables continuous improvement and adaptation of products and services.

NEED OF DATA ANALYTICS

Cost Reduction

- **Fraud Detection:**

- Analytics tools can detect anomalies and patterns that indicate fraudulent activities, saving significant costs associated with fraud.

- **Operational Savings:**

- Identifying and eliminating inefficiencies through data analytics leads to substantial cost savings in operations.

NEED OF DATA ANALYTICS

Improved Marketing and Sales

- **Campaign Effectiveness:**

- Analyzing marketing campaign data helps assess the effectiveness of different strategies and channels, optimizing marketing spend.

- **Sales Forecasting:**

- Data analytics provides accurate sales forecasts, helping organizations manage their sales pipeline and revenue expectations.

NEED OF DATA ANALYTICS

Supply Chain Management

- **Demand Forecasting:**

- Analyzing historical sales and market trends helps predict future demand, leading to better supply chain planning.

- **Inventory Management:**

- Data analytics helps maintain optimal inventory levels, reducing costs associated with overstocking or stockouts.



WALLMART CASE STUDY

CASE STUDY



WALLMART CASE STUDY



11,500 stores in 27 countries, 5,900 retail units, more than 700,000 associates serving more than 100 million customers every week



collects over 2.5 petabytes of data from 1 million customers every hour



In Data Cafe, over 200 streams of internal and external data, including 40 petabytes of recent transactional data, can be modeled, manipulated, and visualized.



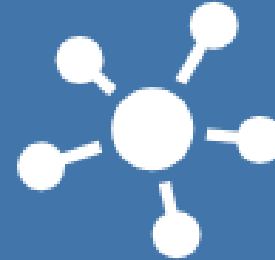
Walmart also constantly analyses over 100 million keywords to know what people near each store are saying on social media. This gives them a better understanding of their customer behavior on what they like and dislike.

DATA ANALYTICS APPLICATION



Healthcare

Banking
sector



Manufacturing

Retail



Life Cycle Phases of Data Analytics

Phase 1: Discovery

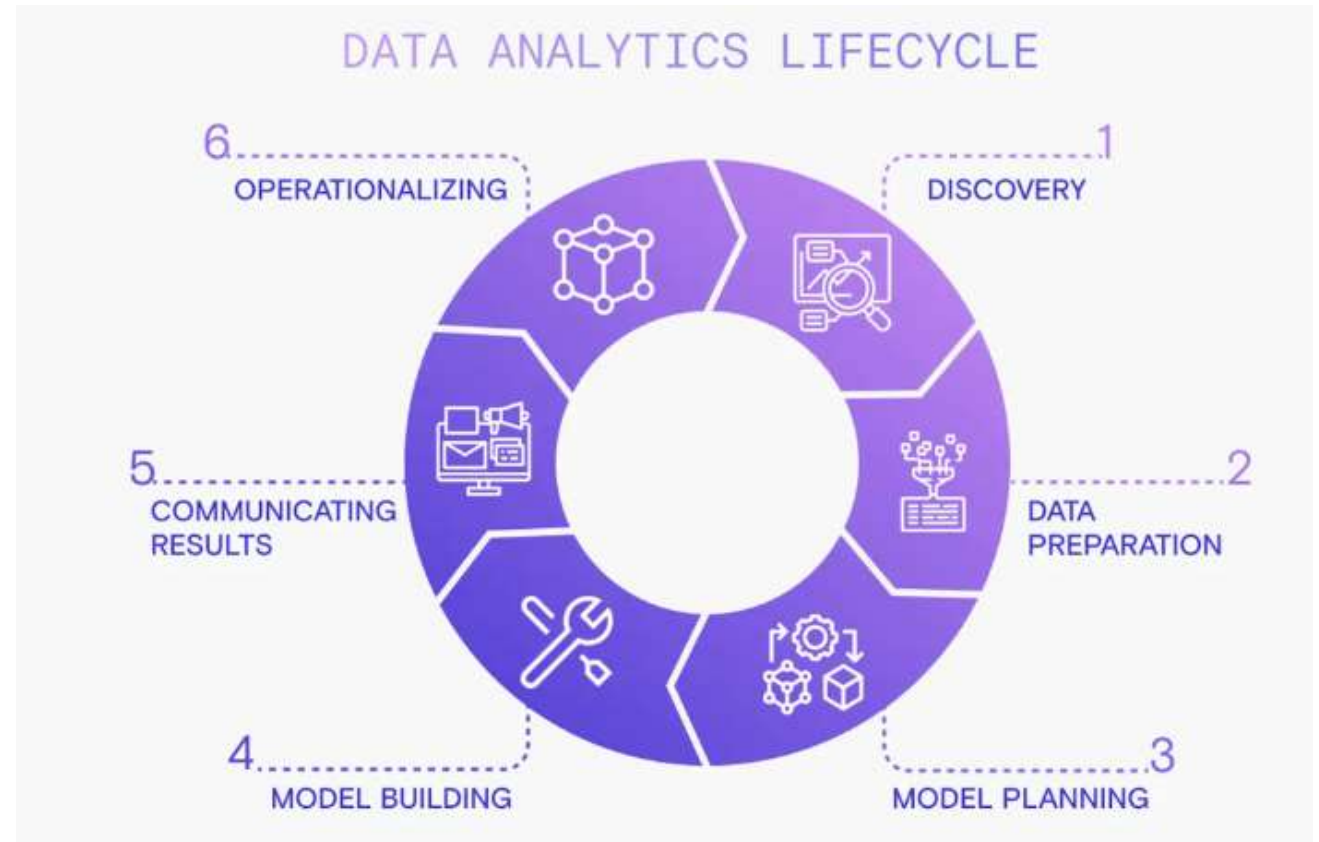
Phase 2: Data Preparation

Phase 3: Model Planning

Phase 4: Model Building

Phase 5: Communication Results

Phase 6: Operationalize



Phase 1: Discovery

- The data science team learns and investigates the problem.
- Develop context and understanding.
- Come to know about data sources needed and available for the project.
- The team formulates the initial hypothesis that can be later tested with data.

Phase 2: Data Preparation

- Steps to explore, preprocess, and condition data before modeling and analysis.
- It requires the presence of an analytic sandbox, the team executes, loads, and transforms, to get data into the sandbox.
- Data preparation tasks are likely to be performed multiple times and not in predefined order.
- Several tools commonly used for this phase are – Hadoop, Alpine Miner, Open Refine, etc.

Phase 3: Model Planning

- The team explores data to learn about relationships between variables and subsequently, selects key variables and the most suitable models.
- In this phase, the data science team develops data sets for training, testing, and production purposes.
- Team builds and executes models based on the work done in the model planning phase.
- Several tools commonly used for this phase are – Matlab and STASTICA.

Phase 4: Model Building

- Team develops datasets for testing, training, and production purposes.
- Team also considers whether its existing tools will suffice for running the models or if they need more robust environment for executing models.
- Free or open-source tools – R and PL/R, Octave, WEKA.
- Commercial tools – Matlab and STASTICA.

Phase 5: Communication Results

- After executing model team need to compare outcomes of modeling to criteria established for success and failure.
- Team considers how best to articulate findings and outcomes to various team members and stakeholders, taking into account warning, assumptions.
- Team should identify key findings, quantify business value, and develop narrative to summarize and convey findings to stakeholders.

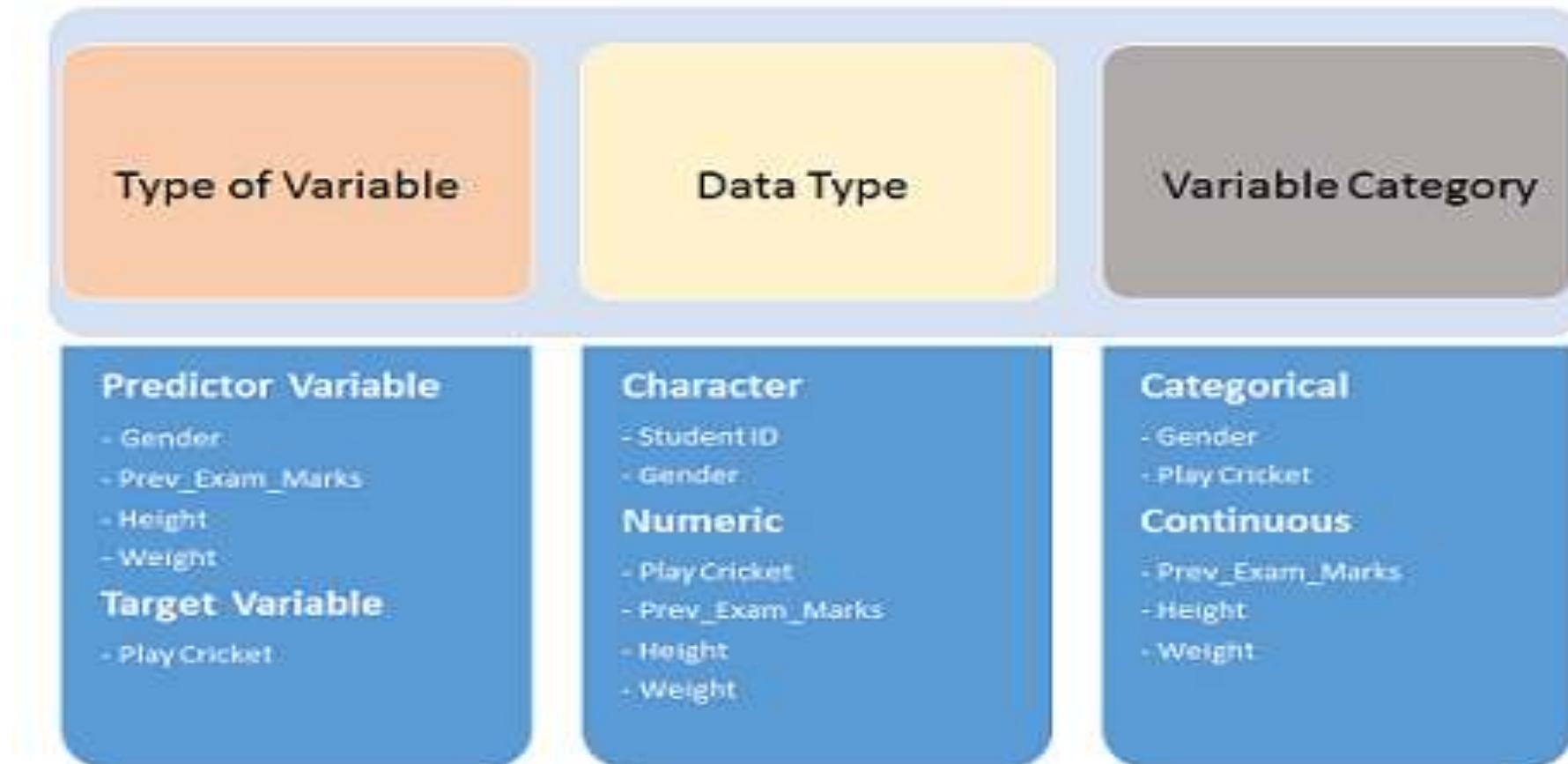
Phase 6: Operationalize

- The team communicates benefits of project more broadly and sets up pilot project to deploy work in controlled way before broadening the work to full enterprise of users.
- This approach enables team to learn about performance and related constraints of the model in production environment on small scale which make adjustments before full deployment.
- The team delivers final reports, briefings, codes.
- Free or open source tools – Octave, WEKA, SQL, MADlib.

Variable Identification in Data Exploration

Student_ID	Gender	Prev_Exam_Marks	Height (cm)	Weight Caregory (kgs)	Play Cricket
S001	M	65	178	61	1
S002	F	75	174	56	0
S003	M	45	163	62	1
S004	M	57	175	70	0
S005	F	59	162	67	0

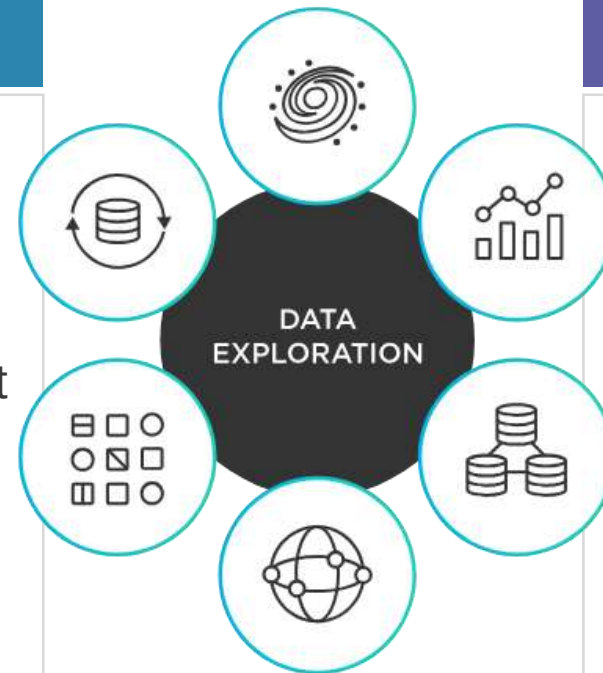
Variable Identification in Data Exploration



Steps of Data Exploration and Preparation

Steps of Data Exploration and Preparation

1. Variable Identification
2. Univariate Analysis
3. Bi-variate Analysis
4. Missing values treatment
5. Outlier treatment
6. Variable transformation
7. Variable creation

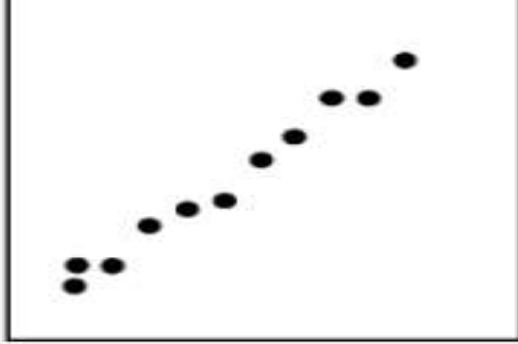


Remember, the quality of your inputs decides the quality of your output

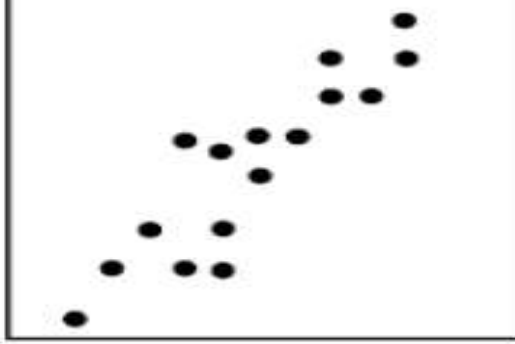
- Data Exploration, Cleaning, And Preparation Can Take Up To 70% Of Your Total Project Time.
- once your business hypothesis is ready, spending a lot of time and effort here makes sense

Bivariate Analysis

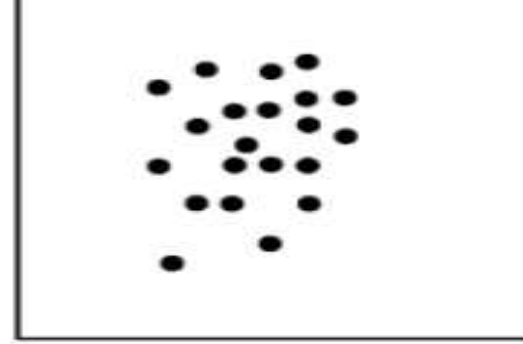
- Bivariate analysis in data exploration refers to finding the relationship between two variables
- We can perform a bivariate analysis combination of
 - Categorical and continuous variables
 - Categorical and categorical
 - Categorical and continuous
 - Continuous and continuous



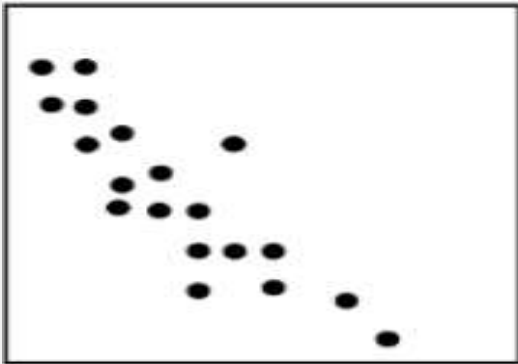
Strong positive correlation



Moderate positive correlation



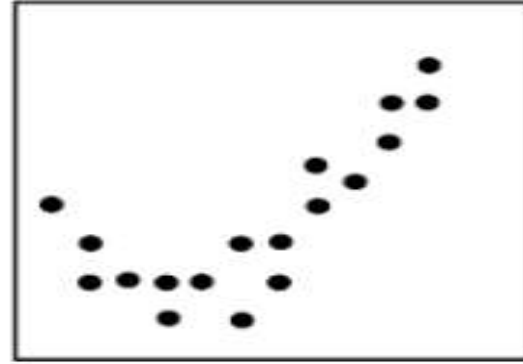
No correlation



Moderate negative correlation



Strong negative correlation



Curvilinear relationship

Continuous & Continuous

Continuous & Continuous

A scatter plot shows the relationship between two variables but does not indicate its strength. To find the strength of the relationship, we use Correlation, which varies between -1 and +1.

- -1: Perfect negative linear correlation
- +1: Perfect positive linear correlation
- 0: No correlation

This correlation can be derived using the following formula:

$$\text{Correlation} = \text{Covariance}(X,Y) / \text{SQRT}(\text{Var}(X) * \text{Var}(Y))$$

X	65	72	78	65	72	70	65	68
Y	72	69	79	69	84	75	60	73

Metrics	Formula	Value
Co-Variance (X,Y)	=COVAR(E6:L6,E7:L7)	18.77
Variance (X)	=VAR.P(E6:L6)	18.48
Variance (Y)	=VAR.P(E7:L7)	45.23
Correlation	=G10/SQRT(G11*G12)	0.65

In above example, we have good positive relationship(0.65) between two variables X and Y

Categorical & Categorical

- **Two-way table**

- We can start analyzing the relationship by creating a two-way table of count and count%.
- The rows represent the category of one variable, and the columns represent the categories of the other variable.
- The count or count% of observations available in each combination of row and column categories.

	Baseball	Basketball	Football	Total
Male	13	15	20	48
Female	23	16	13	52
Total	36	31	33	100

Two-way table

Categorical & Categorical

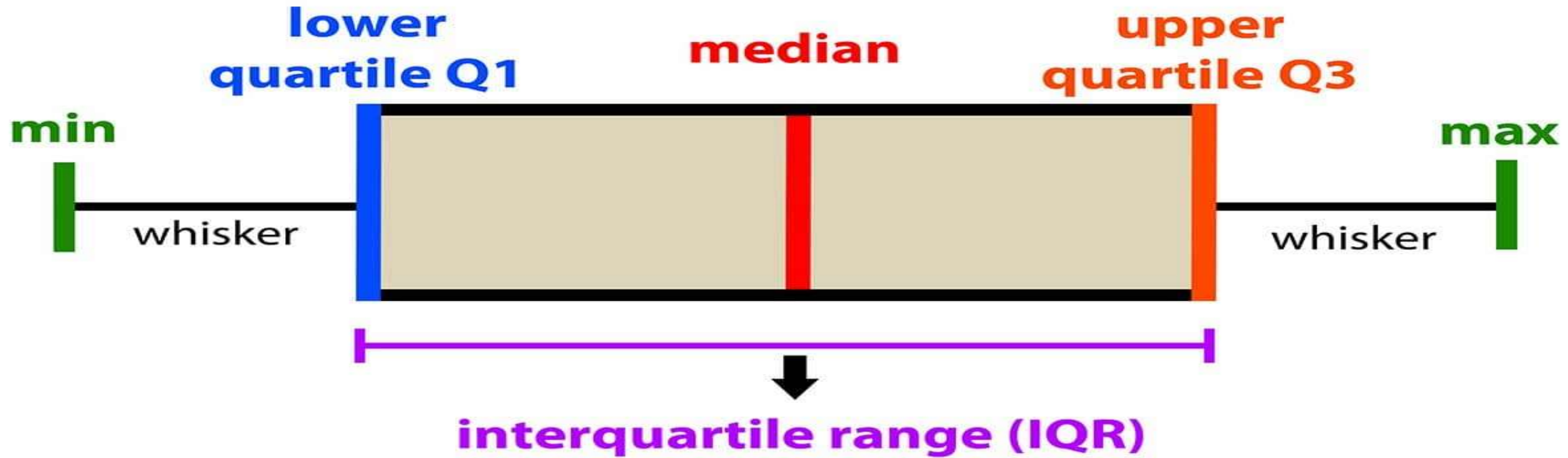
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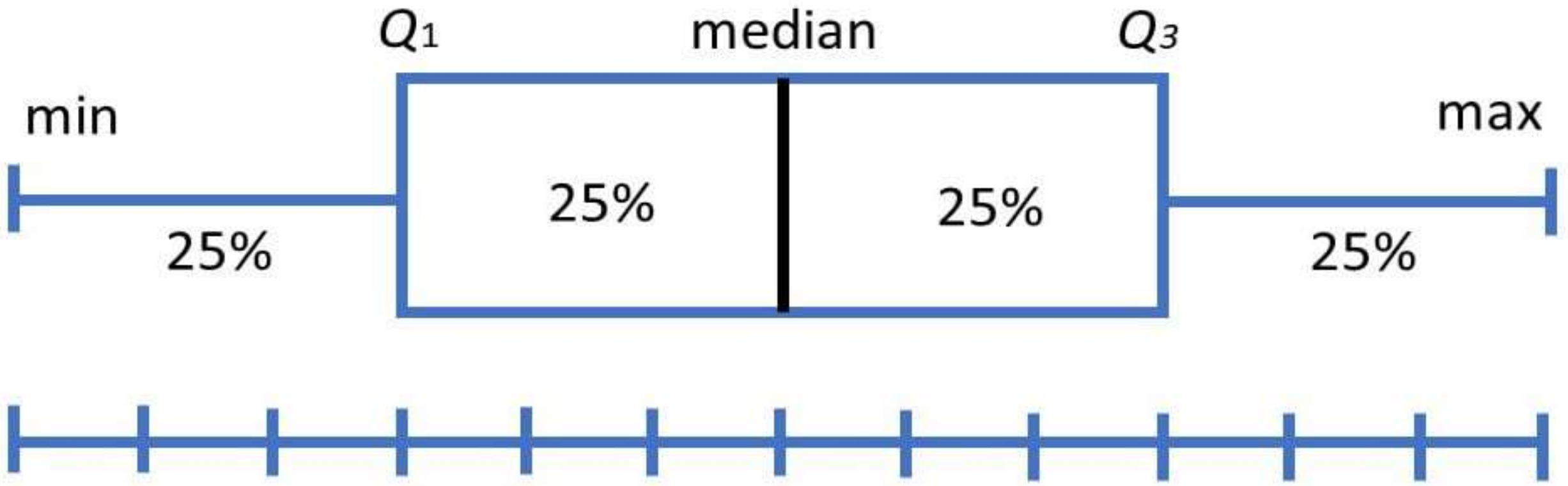
Categorical & Continuous

- To explore the relation between categorical and continuous variables, we can draw box plots for each level of categorical variables

introduction to data analysis: Box Plot



Box Plots



Box Plots

Missing Value Treatment

- To explore the relation between categorical and continuous variables, we can draw box plots for each level of categorical variables

Name	Weight	Gender	Play Cricket/ Not
Mr. Amit	58	M	Y
Mr. Anil	61	M	Y
Miss Swati	58	F	N
Miss Richa	55		Y
Mr. Steve	55	M	N
Miss Reena	64	F	Y
Miss Rashmi	57		Y
Mr. Kunal	57	M	N

Gender	#Students	#Play Cricket	%Play Cricket
F	2	1	50%
M	4	2	50%
Missing	2	2	100%

Name	Weight	Gender	Play Cricket/ Not
Mr. Amit	58	M	Y
Mr. Anil	61	M	Y
Miss Swati	58	F	N
Miss Richa	55	F	Y
Mr. Steve	55	M	N
Miss Reena	64	F	Y
Miss Rashmi	57	F	Y
Mr. Kunal	57	M	N

Gender	#Students	#Play Cricket	%Play Cricket
F	4	3	75%
M	4	2	50%

Why is Missing Values Treatment Required?

Deletion

List wise deletion

Gender	Manpower	Sales
M	25	343
F	.	280
M	33	332
M	.	272
F	25	.
M	29	326
.	26	259
M	32	297

Pair wise deletion

Gender	Manpower	Sales
M	25	343
F	.	280
M	33	332
M	.	272
F	25	.
M	29	326
.	26	259
M	32	297

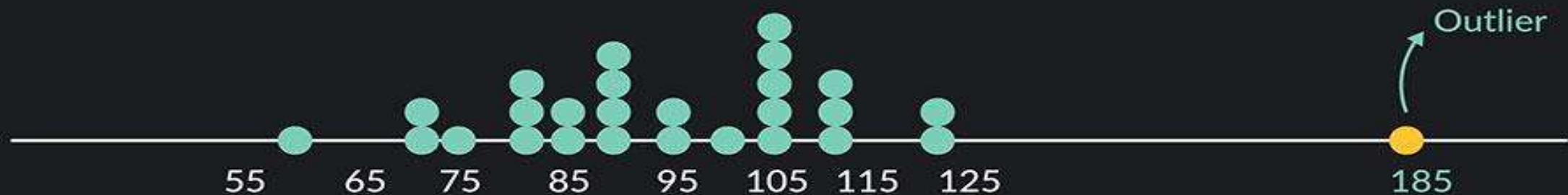
- It is of two types: List Wise Deletion and Pair Wise Deletion.
- In list-wise deletion, we delete observations where any variable is missing. Simplicity is one major advantage of this method, but this method reduces the power of the model because it reduces the sample size.
- In pair-wise deletion, we analyze all cases where the variables of interest are present. This method has the advantage of keeping as many cases available for analysis as possible. One disadvantage is that it uses different sample sizes for other variables.

Other Methods For Missing Value Treatment

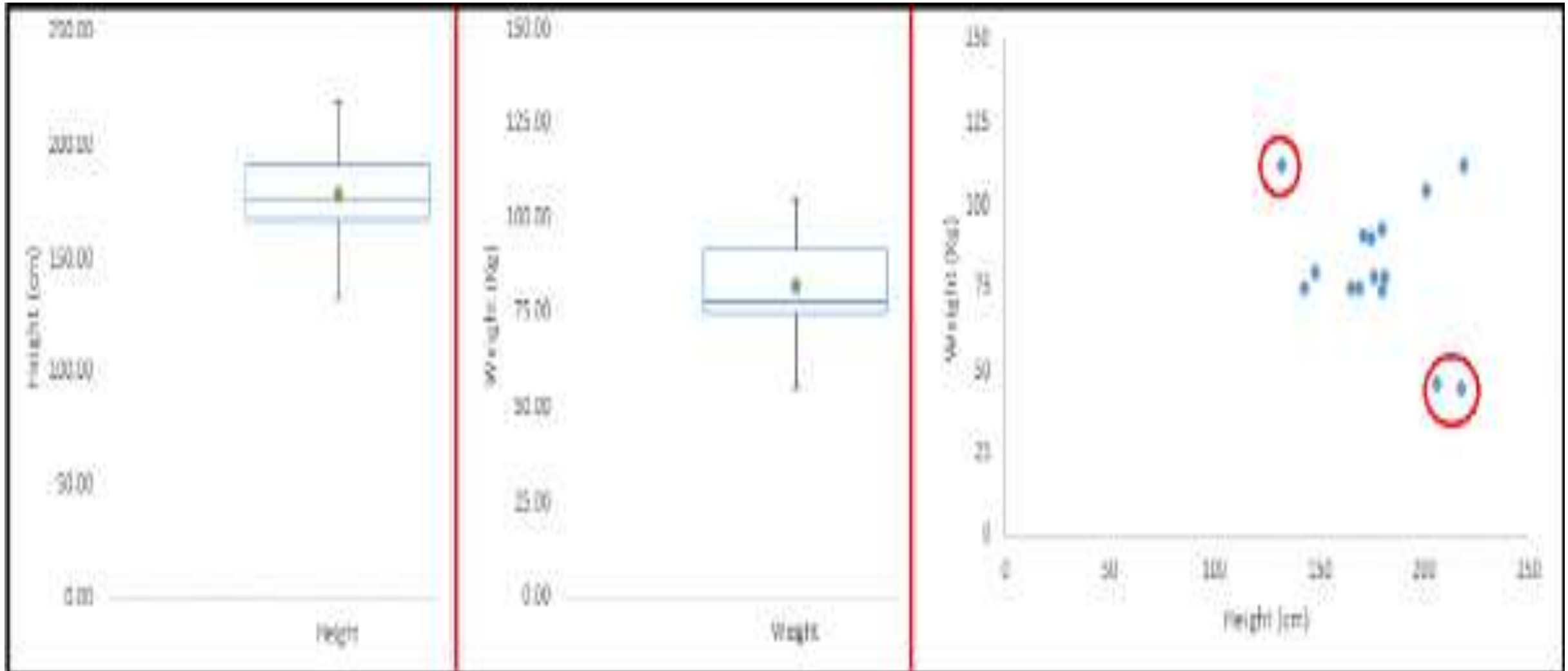
- Mean/ Mode/ Median Imputation
- Similar Case Imputation
- Prediction Model

An Outlier Is A Data Point That Differs Significantly From Other Observations

What is an **Outlier**
and How to Find it?



Univariate and Multivariate Outliers



Types of Outliers

- Data Entry Errors
- Measurement Error
- Experimental Error
- Intentional Outlier
- Data Processing Error
- Sampling error

Without Outlier

4, 4, 5, 5, 5, 5, 6, 6, 6, 7, 7

Mean = 5.45

Median = 5.00

Mode = 5.00

Standard Deviation = 1.04

With Outlier

4, 4, 5, 5, 5, 5, 6, 6, 6, 7, 7, 300

Mean = 30.00

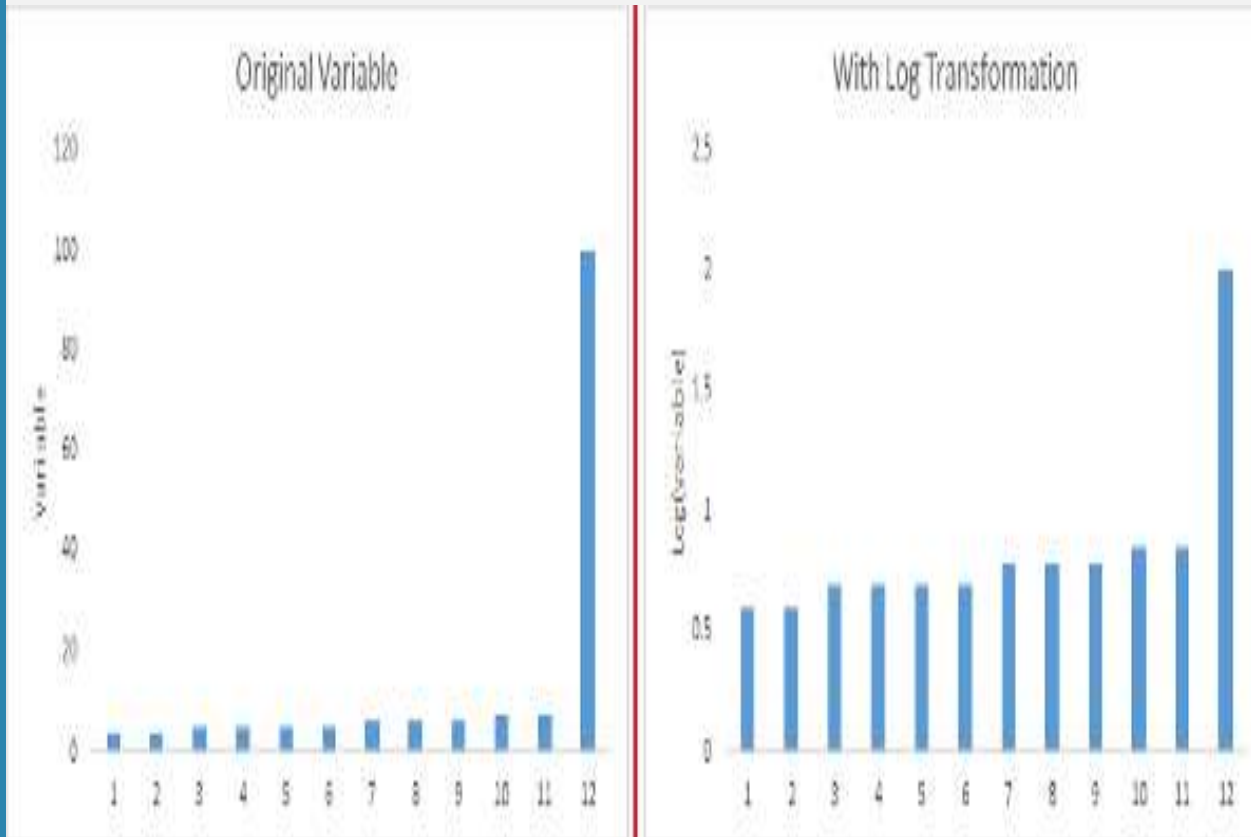
Median = 5.50

Mode = 5.00

Standard Deviation = 85.03

What is the Impact of Outliers on a Dataset?

How to Remove Outliers?



- 1. Deleting observations**
- 2. Transforming and binning**
- 3. Imputing**
- 4. Treat separately**

What is Variable Transformation?

- In data modeling, transformation refers to replacing a variable with a function
- For instance, replacing a variable x by the square/cube root or logarithm x is a transformation. In other words, transformation is a process that changes the distribution or relationship of a variable with others

What is Feature / Variable Creation & What Are Its Benefits?

Emp_Code	Gender	Date	New_Day	New_Month	New_Year
A001	Male	21-Sep-11	21	9	2011
A002	Female	27-Feb-13	27	2	2013
A003	Female	14-Nov-12	14	11	2012
A004	Male	07-Apr-13	7	4	2013
A005	Female	21-Jan-11	21	1	2011
A006	Male	26-Apr-13	26	4	2013
A007	Male	15-Mar-12	15	3	2012

What is Feature / Variable Creation & What Are Its Benefits?

Emp_Code	Gender	Var_Male	Var_Female
A001	Male	1	0
A002	Female	0	1
A003	Female	0	1
A004	Male	1	0
A005	Female	0	1
A006	Male	1	0
A007	Male	1	0

Big Data Phase 1 – Structured Content	Big Data Phase 2 – Web Based Unstructured Content	Big Data Phase 3 – Mobile and Sensor Based Content
Period: 1970-2000	Period: 2000 – 2010	Period: 2010 - Present
• RDBMS & data warehousing • Extract Transfer Load • Online Analytical Processing • Dashboards & scorecards • Data mining & statistical analysis	• Information retrieval and extraction • Opinion mining • Question answering • Web analytics and web intelligence • Social media analytics • Social network analysis • Spatial-temporal analysis	• Location-aware analysis • Person-centred analysis • Context-relevant analysis • Mobile visualization • Human-Computer-Interaction

EVOLUTION OF BIG DATA

7 Essential Big Data Best Practices

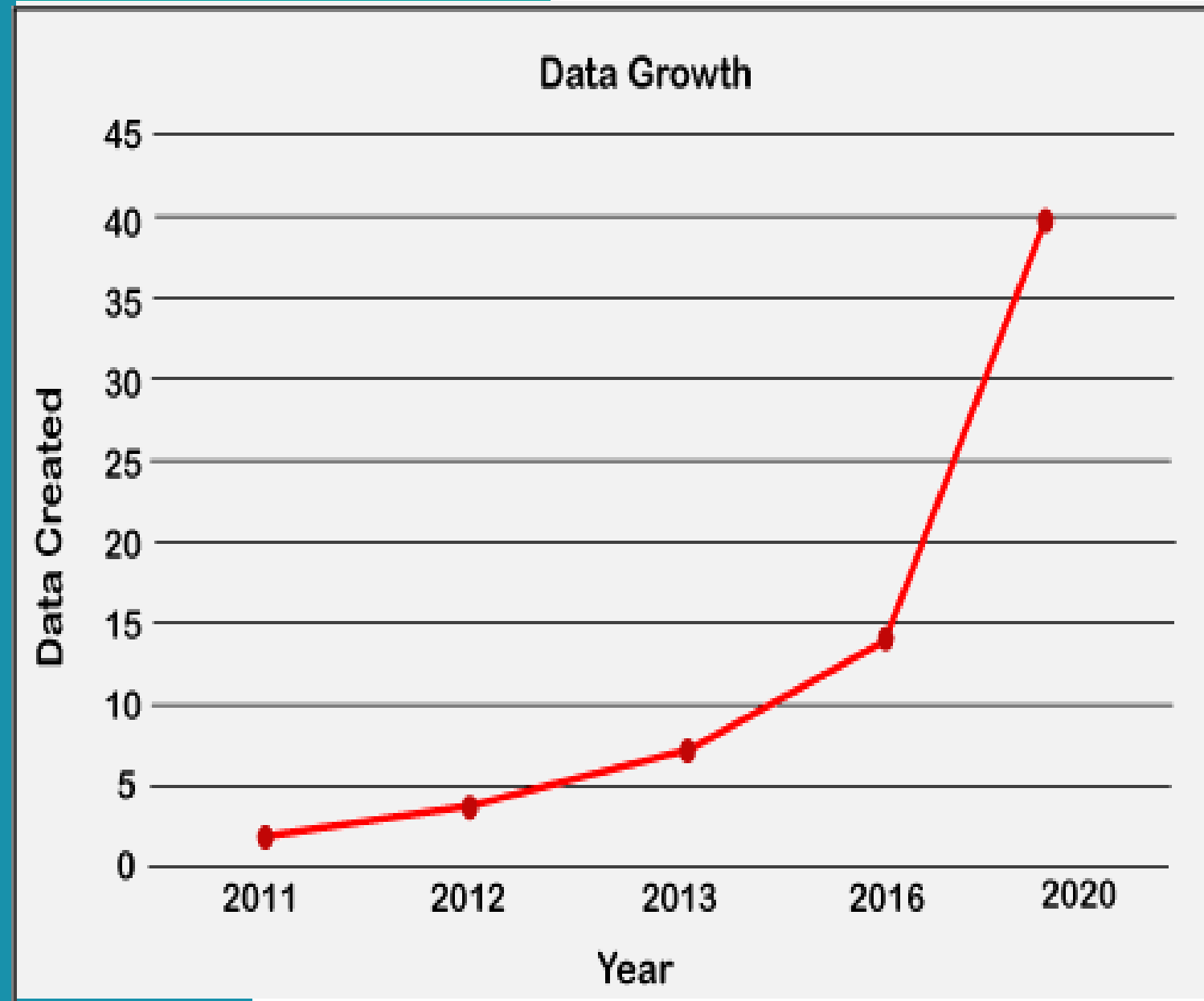
- **Implement Data Quality Management Programs**
- **Build More Scalable Infrastructures**
- **Employ Agile Development Methodologies**
- **Safeguard Data With Robust Security Measures**
- **Use Data Ethically**
- **Monitor And Optimize Continuously**
- **Provide Workforce Skill Development**

BIG DATA CHARACTERISTICS



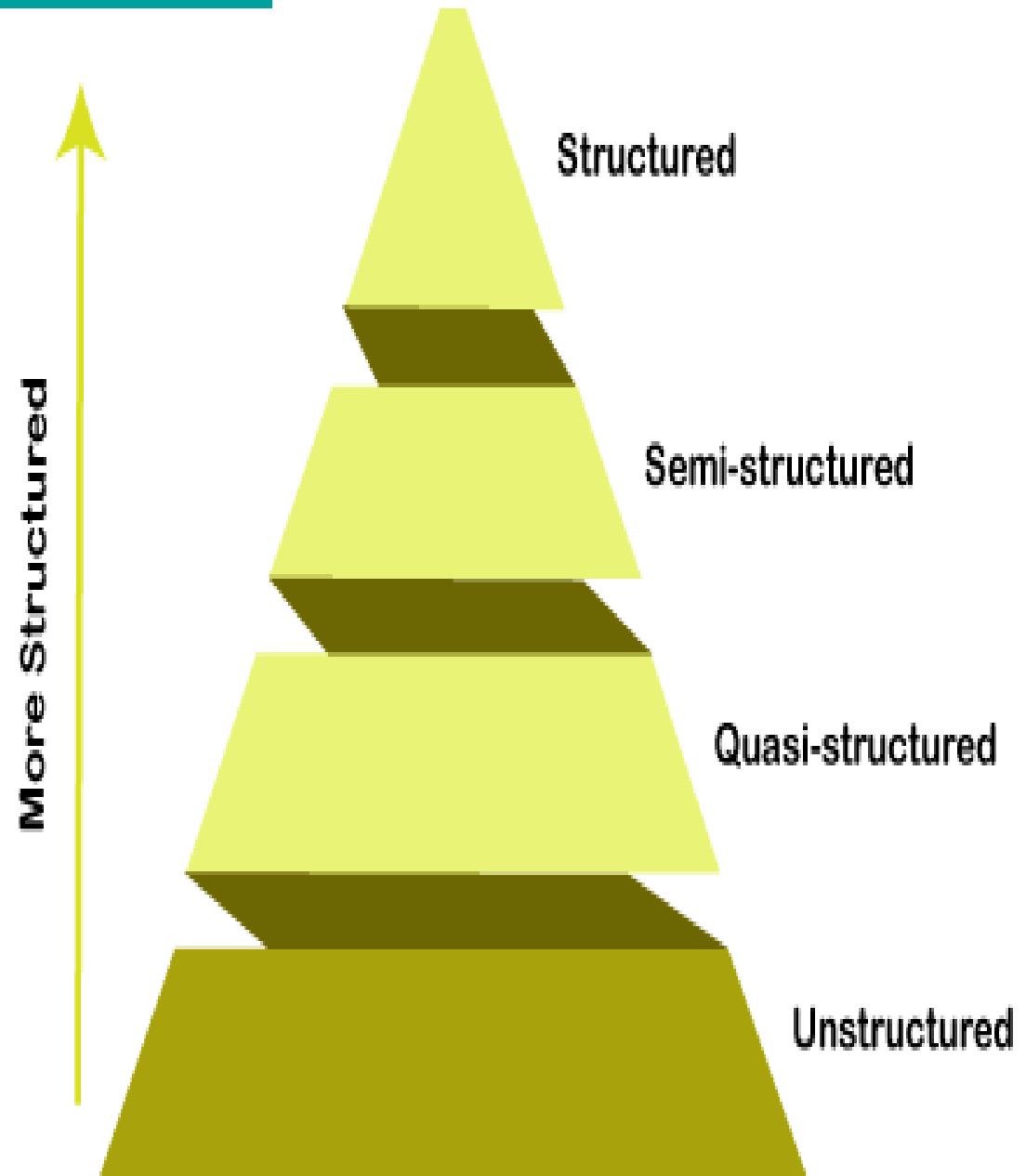
Volume

- Big Data is a vast 'volumes' of data generated from many sources daily, such as business processes, machines, social media platforms, networks, human interactions, and many more.
- Facebook can generate approximately a billion messages, 4.5 billion times that the "Like" button is recorded, and more than 350 million new posts are uploaded each day. Big data technologies can handle large amounts of data.



Variety

- Big Data can be structured, unstructured, and semi-structured that are being collected from different sources.
- Data will only be collected from databases and sheets in the past, But these days the data will comes in array forms, that are PDFs, Emails, audios, SM posts, photos, videos, etc.



Veracity

- Veracity basically means the degree of reliability that the data has to offer.
- Since a major part of the data is unstructured and irrelevant, Big Data needs to find an alternate way to filter them or to translate them

Value

It is actually the amount of valuable, reliable and trustworthy data that needs to be stored, processed, analyzed to find insights

Velocity

- Velocity creates the speed by which the data is created in real-time.
- It contains the linking of incoming data sets speeds, rate of change, and activity bursts.
- The primary aspect of Big Data is to provide demanding data rapidly

